

Kundan Kumar

Lead Backend Engineer — 8.5 Years Experience

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EDUCATION

- **Indian Institute of Technology, Delhi** Delhi, IN
Bachelor of Technology in Electrical and Electronics Engineering; AIR: 1559 (Category Rank: 174) Aug 2010 – Jun 2014

EXPERIENCE

- **Leadbeam, Lead Backend Engineer** Bangalore, May 2025 – Present
 - **Lead Scoring:** Architected a versioned scoring system over a ~15M-business dataset combining ~50 ICP signals (firmographics, web presence, vertical tags), time-decaying engagement momentum, and configurable boost rules; PostGIS geo-bounding scores ~1M leads per company across 100+ independent configs in Celery batches.
 - **Data Pipeline:** Designed the Atlas ingestion pipeline that consolidates 15M+ US business records from Overture Maps, Google Places, OpenWebNinja, OSM, and CSV sources into a unified golden-record store with multi-signal deduplication, field-level provenance, and confidence-weighted merging — powering SMB lead discovery.
 - **AI Framework & Evaluation:** Built a unified AI execution framework with versioned prompts/schemas, idempotent runs, per-attempt token & cost tracking, and full audit trails, plus a schema-driven evaluation layer (JSON Schema validation, classified failure modes, auto-retry, usage analytics) — powering and monitoring all AI tasks across the product.
 - **AI Error Simplification:** Built a RAG pipeline that turns cryptic backend errors into clear user-facing messages — OpenAI embeddings + pgvector semantic search retrieve similar past errors to ground LLM generation, backed by a semantic cache that serves repeat errors with zero LLM calls; sub-ms hot path with async Celery workers.
 - **Observability:** Introduced structured JSONL logging and monitoring with request-id tracing that propagates across background threads and thread pools, cutting log aggregation and incident triage to within minutes.
- **Pasarpolis, Engineering Lead** Gurgaon, May 2023 – May 2025
 - **Product Expansion & Team Leading:** Owned the Agency vertical with a team of 7, architecting and onboarding 4 major products end-to-end — Travel Protection, Extended Warranty, Offline-to-Online flow, and an MLM performance system.
 - **Infrastructure Cleanup:** Merged redundant services (Gateway, Payment, Policy Creation) by enabling dual writes and ensuring data consistency, reducing engineering overhead and infra complexity.
 - **Wallet Consistency:** Resolved wallet-service concurrency issues and boosted throughput with database row locks and Redis distributed locks, then split the single bottleneck ledger account into 10 partitioned wallets with periodic background rebalancing to spread write contention.
 - **Support Automation:** Introduced internal OPS tools and enhancements based on ticket trend analysis, reducing recurring support tickets by ~90%.
 - **Auth Optimization:** Transitioned user login from OTP to PIN-based flow across mobile/web platforms to improve UX and reduce authentication time.
- **Pasarpolis, Software Engineer** Gurgaon, Dec 2020 – Mar 2022
 - **Core Service Development:** Built agent-facing services to compare and book insurance policies using Django, Redis, Kafka, MySQL, and MongoDB — all hosted on GCP.
 - **Financial Modules:** Designed modular financial components — tax calculation, policy endorsements, invoice generation, and a pluggable commission service — handling sensitive transaction flows.
- **Nykaa, Senior Software Engineer** Gurgaon, Mar 2022 – Apr 2023
 - **System Optimization:** Implemented application-side caching to reduce Redis network calls by ~70% and cut API latency for the revenue attribution service.
 - **High-Traffic Platform & Data Pipeline:** Built backend services powering dynamic widgets and personalized experiences across millions of daily requests on scalable NoSQL systems; redesigned the revenue attribution stream from Pub/Sub to a cron-based Airflow pipeline feeding JSON deltas to data science via PySpark for transformation and billing.

— 3-year career break to pursue UPSC, reaching final-stage interviews —
- **Amazon, SDE-1** Chennai, Aug 2017 – Feb 2018
 - **Infra Cost Reduction:** Refactored Gateway service into a reusable internal library for downstream teams, projected to save ~\$4M annually in infrastructure costs.
 - **Service Ownership:** Maintained Gateway and Search & Discovery services for Amazon Appstore, ensuring performance and scalability across global regions.
- **Robert Bosch, Senior Engineer** Bangalore, Jul 2014 – Jun 2017
 - **Embedded Systems R&D:** Led 0-to-1 development of Bluetooth tyre-pressure monitoring, Wi-Fi blind-spot camera, and Zigbee-based smart-field systems for automotive and IoT pilots.
 - **Interface Libraries:** Built reusable dot-matrix LCD display libraries enabling alphanumeric input and menu rendering via navigation keys.
 - **Medical Device:** Architected a handheld blood potassium monitoring device end-to-end, owning embedded firmware and hardware design.

TECHNICAL AND PROGRAMMING SKILLS

- Python, Java, C, SQL, NoSQL, AWS, GCP, Django, Flask, Spring, Kafka, Redis, Celery, Docker, PySpark, React, RAG, LLMs, LangChain, LangGraph, Gen AI, Agentic AI, Agentic Workflows, SOLID Principles, Microservices